International Journal of Computing and Digital Systems
2025, VOL. 18, NO. 1, 1–13

http://dx.doi.org/10.12785/ijcds/XXXXXX

ProTrader AI: A Multi-Source Hybrid Ensemble with LLM-Augmented Sentiment, Conformal Uncertainty, and Explainable AI for Indian Equity Market Prediction and Cross-Market Transfer

Anandkumar Pardeshi[1] **and Sujata Deshmukh**[2]

[1] *Department of Computer Science and Engineering, Fr. C. Rodrigues Institute of Technology, Vashi, India*
[2] *Department of Computer Engineering, Fr. C. Rodrigues College of Engineering, Bandra, India*

Received 15 January 2025, Revised 20 March 2025, Accepted 25 April 2025

Abstract: Equity price forecasting in emerging markets such as India is complicated by multilingual news flows, regime changes, structural illiquidity, and a heterogeneous mix of retail and institutional participants. We present *ProTrader AI*, a hybrid ensemble with an open-source implementation that fuses four data streams—Open-High-Low-Close-Volume (OHLCV) time series, multi-source news and social-media sentiment scored by a domain-adapted FinBERT model with multilingual support, Foreign and Domestic Institutional Investor (FII/DII) flows, and macroeconomic state variables—into a Ridge-stacked classifier and a conformal quantile regressor. The system is explicitly designed around four research objectives: (RO1) comprehensive multi-source data integration; (RO2) a hybrid large-language-model (LLM) plus indicator framework for real-time prediction; (RO3) an explainable AI layer with temporal sentiment-to-price dynamics analysis; and (RO4) cross-market transfer learning evaluation. On a 2024 holdout of NIFTY 50 constituents, ProTrader AI achieves **59.3%** directional accuracy (95% block-bootstrap CI [56.1%, 62.7%]; block length 5, reflecting cross-sectional dependence across 50 tickers), Expected Calibration Error (ECE) of **4.2%**, Brier score **0.218**, and Sharpe ratio **1.34**, with a Diebold–Mariano $p = 0.031$ against buy-and-hold. XGBoost-component SHAP attribution identifies FinBERT sentiment as the single most influential feature (mean absolute SHAP value $|\bar{\phi}| = 0.182$), and Granger-causality testing reveals a significant one-day sentiment lead ($p = 0.018$). A zero-shot cross-market transfer to an S&P 500 subset yields 55.8% accuracy, rising to 57.6% after a lightweight 30-day fine-tune of the stacking layer only.

Keywords: stock market prediction; multi-source data fusion; FinBERT; ensemble learning; conformal prediction; SHAP; transfer learning; Indian equity markets; explainable AI.

1. INTRODUCTION

The application of artificial intelligence to financial decision-making has accelerated dramatically, driven by the availability of high-frequency market data, alternative data streams, and large pre-trained language models [1]. Yet, despite measurable progress, three persistent gaps remain: (i) most published systems fuse at most two data modalities, ignoring the complementary signal carried by macroeconomic state and institutional order-flow; (ii) probabilistic outputs are rarely calibrated, making the resulting confidence scores unsafe for risk-proportional position sizing; and (iii) very few studies validate cross-market generalisation, leaving open whether a model trained on one exchange transfers to another.

The Indian National Stock Exchange (NSE) presents a particularly demanding testbed. It hosts approximately 2,000 actively traded equities and combines foreign institutional investors (FIIs), domestic institutions (DIIs), and a rapidly expanding retail base whose decisions are shaped by news-app feeds in both English and Hindi. Nti et al. [2] and Long et al. [3] have independently shown that fusing heterogeneous data modalities lifts accuracy above any single-source baseline, but neither work addresses the Indian institutional-flow channel, multilingual sentiment noise, or calibrated uncertainty.

E-mail address: anand.pardeshi@fcrit.ac.in

This paper introduces *ProTrader AI*, a hybrid ensemble specifically engineered for NSE equities and evaluated for cross-market transfer to the U.S. S&P 500. Our work is organised around four explicit research objectives:

- **RO1 — Multi-Source Integration:** Investigate a comprehensive model integrating diverse data sources—financial metrics, news articles, social-media sentiment, and macroeconomic factors—for holistic stock market prediction.
- **RO2 — Hybrid LLM + Indicator Framework:** Design a hybrid framework combining large language models, sentiment-analysis techniques, and conventional financial indicators, focusing on optimising deep-learning architectures for real-time stock market predictions.
- **RO3 — Explainability and Temporal Sentiment Dynamics:** Establish an explainable AI framework capable of delivering transparent and interpretable predictions, analyse the impact of sentiment-data noise, and explore temporal dynamics between sentiment shifts and stock price movements.
- **RO4 — Cross-Market Evaluation and Transfer Learning:** Evaluate and validate the proposed model across different stock markets, explore transfer-learning techniques for cross-market applications, and validate against benchmark datasets and real-world market data.

The remainder of the paper is structured to address these objectives in sequence. Section 2 reviews related work in four subsections aligned with RO1–RO4. Section 3 formalises the methodology with all mathematical details. Section 4 describes the experimental protocol. Section 5 presents empirical results, including cross-market transfer and explainability analysis. Section 6 discusses each RO against the evidence, and Section 7 concludes with a future-work roadmap.

2. RELATED WORK

A. Multi-Source Data Fusion for Stock Prediction (RO1)

The case for fusing heterogeneous information has been made repeatedly. Nti et al. [2] proposed a deep neural fusion framework combining technical, fundamental, and macroeconomic streams, reporting a measurable accuracy uplift over single-source baselines. Long et al. [3] extended the idea to multi-view heterogeneous data—graph-structured news plus tabular price—and demonstrated that view-level attention outperforms naive feature concatenation. Fu and Zhang [4] integrated investor sentiment from social media with traditional price features, showing that the marginal information content of sentiment is non-redundant with respect to technical indicators.

A recurring methodological theme in this body of work is that the order in which heterogeneous streams are fused—feature-level concatenation, view-level attention, or late-stage ensembling—materially shapes the achievable generalisation. Long et al. [3] report that view-level attention matters most when one stream (news) has a sharply different sampling cadence than the others (price), a setting that mirrors the Indian market where institutional flows and macro variables update at coarser intervals than intraday OHLCV. Conversely, naive concatenation overfits to the highest-cadence stream, an effect Fu and Zhang [4] attribute to gradient imbalance during deep-network training. ProTrader AI sidesteps this by stacking heterogeneous base learners, each specialised on a subset of streams, and recombining them via a Ridge meta-learner—a late-fusion design that respects each stream's natural cadence.

A second under-explored axis is regional specificity. None of the cited multi-source frameworks evaluate on emerging-market exchanges, where microstructural noise (illiquidity, circuit-breakers), bilingual news flows, and the FII/DII institutional channel jointly shape price discovery. Mu et al. [5] note that investor-sentiment effects are markedly stronger in markets with high retail participation—a structural feature of the NSE. ProTrader AI extends this paradigm by adding (a) FII/DII institutional order-flow, which is unique to the Indian market, and (b) macroeconomic state variables (VIX, INR/USD, crude, gold) as conditioning context—sources not considered jointly in any prior multi-source system for Indian equities.

B. Hybrid Models Combining Sentiment and Technical Indicators (RO2)

Hybridisation of sentiment with classical indicators has been explored in various forms. Moodi et al. [6] fused FinBERT-derived sentiment with technical indicators inside a deep-learning framework and demonstrated reduced prediction regret in volatile periods. Farhadi et al. [7] proposed a hybrid LSTM–GRU model that exploits complementary inductive biases of the two recurrent architectures. Sawhney et al. [8] combined text, audio, and price signals in a multi-modal graph neural network for stock movement prediction, demonstrating the value of fusing modalities that capture complementary information horizons. Both papers treat the language model purely as a feature extractor without probability calibration, conformal uncertainty, or institutional-flow integration—all of which ProTrader AI adds. On the evaluation of sentiment models for finance, Mishev et al. [9] benchmarked the full lexicon-to-transformer evolution and established FinBERT as the de-facto choice for English-language financial NLP; Yang et al. [10] proposed an independent FinBERT variant, further motivating the adoption of domain-adapted BERT architectures here. Liu et al. [11] reviewed the rise of LLM-based sentiment systems and noted that interpretability is the most-cited blocker to adoption in production quantitative systems.

A persistent weakness of the recurrent-network branch of this literature is the lack of out-of-fold (OOF) discipline

E-mail address: anand.pardeshi@fcrit.ac.in

when stacking is attempted. Many hybrid LSTM-FinBERT systems train all components on the same temporal window, allowing the meta-learner to glimpse signals that base learners have already seen, which inflates reported accuracy by 2–4 percentage points in our reproductions. ProTrader AI's expanding-window cross-validation makes leakage structurally impossible—a methodological correction we view as a prerequisite for credible reporting. The complementary-architecture argument of Farhadi et al. [7] is preserved here through our heterogeneous base-learner pool (XGBoost, LightGBM, GRU, ARIMA, Prophet), which combines tree-based, recurrent, and classical-statistical inductive biases under a common stacker.

The third recurring concern in this subgroup is the gap between predictive accuracy and tradeable Sharpe. Moodi et al. [6] report a hybrid accuracy gain of 4–5 pp but do not couple this to risk-adjusted returns under realistic frictions. Liu et al. [11] argue—and our empirical evidence corroborates—that uncalibrated probabilities, when fed to a Kelly-like sizing rule, can erase the entire accuracy edge through over-confident position sizing. ProTrader AI directly addresses this by applying isotonic calibration on out-of-fold residuals before any sizing decision is made, ensuring that confidence scores are interpretable as empirical hit-rates rather than abstract logits.

C. Large Language Models, Explainability, and Temporal Dynamics (RO3)

Guo et al. [12] articulate a "Quant 4.0" vision requiring automated, explainable, and knowledge-driven AI, and call for SHAP-style attribution as a minimum reporting standard in quantitative finance. We adopt this mandate and contribute a Granger-causality treatment of the temporal sentiment-to-price coupling. Ónozó et al. [13] showed that LLMs can nowcast macroeconomic indicators from financial news, establishing a complementary path to the structured feature approach we use. Parekh et al. [14] demonstrated that deep learning combined with financial-domain sentiment generalises to cryptocurrency markets, suggesting robustness across asset classes.

A subtle but consequential limitation of prior LLM-finance studies is the implicit assumption that sentiment information is contemporaneous with price. The Granger-causality approach we adopt directly tests the temporal *lead* of sentiment over returns, distinguishing between predictive information and post-hoc rationalisation. Empirically this matters: news that arrives after the close drives the next session's open rather than the current close, and our sentiment-aggregation logic respects this boundary by routing post-15:30 IST headlines into the next trading day's feature vector. The investor-sentiment conditioning of Mu et al. [5] further motivates our explicit inclusion of FinBERT polarity as a first-class feature rather than a post-hoc filter.

A second methodological issue is the granularity of explanation that SHAP-on-deep-network systems can provide. While Lundberg and Lee's unified game-theoretic framework [15] is universally applicable, the variance of attribution estimates on small validation sets can be substantial. We address this by computing TreeSHAP on the XGBoost base learner—where exact Shapley values are tractable in polynomial time—and propagating them to the stacker through Ridge coefficients. This preserves both interpretability and computational tractability, an engineering trade-off that mirrors the production-readiness concerns flagged by Liu et al. [11] as the principal blocker to LLM adoption in quantitative finance.

D. Cross-Market Transfer Learning (RO4)

Cross-market generalisation has the thinnest empirical record. Li et al. [16] pioneered "sentimental transfer learning" from a source market with rich annotation to a target with sparse labels, reporting gains under domain adaptation. Kim et al. [17] used effective transfer entropy as a non-parametric tool for measuring inter-market information flow on U.S. equities, finding that information leadership persists across different exchange windows. Lu et al. [18] addressed the heterogeneity between machine and human information processing in stock prediction, calling for models that can bridge diverse market micro-structures—a property our lightweight adapter-layer design supports.

A practical critique of full-network domain-adaptation methods is their data hunger: re-fitting hundreds of millions of LLM parameters on a target market demands either annotated corpora or substantial compute. Our two-stage adapter design—freezing the FinBERT encoder and the heterogeneous base learners while re-fitting only the Ridge stacker and isotonic calibrator (< 1,000 parameters)—is deliberately engineered to be parameter-efficient, echoing the recent surge in adapter-based methods in NLP. The empirical result that 30 days of target-market data recover 1.8 pp of the zero-shot deficit suggests that the bottleneck in cross-market transfer is recombination, not representation.

A separate but related question is whether information leadership is symmetric across markets. Kim et al. [17] report asymmetric transfer-entropy patterns between the U.S. and several Asian markets, with the U.S. leading at short lags and trailing at longer ones. Our zero-shot 55.8% accuracy on S&P 500—which exceeds buy-and-hold by 4.6 pp—suggests that an Indian-trained ProTrader AI captures generalisable equity-market structure rather than India-specific idiosyncrasies. Our cross-market experiment directly extends these works: we evaluate both zero-shot and 30-day fine-tuned transfer of a NIFTY-trained ProTrader AI to an S&P 500 subset, providing an early systematic benchmark of multi-source emerging-to-developed market transfer on the NSE/S&P 500 pair.

3. METHODOLOGY

The ProTrader AI pipeline is structured in five stages illustrated in Figure 1: (A) multi-source data ingestion, (B) feature engineering, (C) base-learner ensemble and stacking, (D) calibration and conformal intervals, and (E) explainability and temporal analysis. Stages (A)–(B)

E-mail address: anand.pardeshi@fcrit.ac.in

address RO1; (B)–(D) address RO2; (E) addresses RO3; and a dedicated cross-market adapter addresses RO4.

A. Multi-Source Data Ingestion (RO1)

Let t index NSE trading days. Four data streams are synchronised to a common daily grid:

Stream 1 (Price/Volume). OHLCV bars for each NIFTY 50 constituent are pulled via *yfinance*. Corporate-action adjustments are inherited from Yahoo Finance.

Stream 2 (News and Social Media). Headlines are aggregated from five sources: NewsAPI (keyed), Google News RSS (English-India locale), Google News RSS (Hindi-India locale), the GNews PyPI package (7-day window, EN + HI), and Economic Times RSS. Articles arriving after 15:30 IST are aligned to the next trading session. Deduplication is performed by 50-character title prefix.

Stream 3 (Institutional Flows). FII and DII daily net cash flows (in INR Cr) are scraped from NSE bhavcopy. Flows are released end-of-day and thus enter the model with a one-day lag to preclude look-ahead bias.

Stream 4 (Macroeconomic State). Four macro proxies are retrieved via yfinance: India VIX (`^INDIAVIX`), INR/USD spot (`INR=X`), front-month Brent crude (`CL=F`), and gold spot (`GC=F`). Their 1-day and 5-day log-returns are clipped to $[-0.20, +0.20]$ and $[-0.40, +0.40]$ respectively to suppress data-glitch outliers.

B. Feature Engineering (RO1, RO2)

From the four streams we compute a joint feature vector $\mathbf{x}_t \in \mathbb{R}^d$ ($d = 36$ including institutional and macro dimensions). The 27 price-volume technical features are detailed in Table I and grouped as follows.

Returns. Log return $r_t = \ln(C_t/C_{t-1})$ and multi-horizon variants $r_t^{(h)} = \ln(C_t/C_{t-h})$ for $h \in \{2, 5, 10, 20\}$.

Momentum oscillators. The 14-period Wilder RSI is

$$\mathrm{RSI}_{14}(t) = 100 - \frac{100}{1 + \mathrm{RS}_{14}(t)}, \quad \mathrm{RS}_{14}(t) = \frac{\overline{U}_{14}(t)}{\overline{D}_{14}(t)}, \tag{1}$$

normalised to $[0, 1]$. We further compute 12/26/9-period MACD signal and histogram (each divided by C_t for stationarity), 14-period Williams %R, and Bollinger Band %B with a 20-period, 2σ envelope:

$$\mathrm{BB}_{\%B}(t) = \frac{C_t - \mathrm{LB}(t)}{\mathrm{UB}(t) - \mathrm{LB}(t) + \varepsilon}. \tag{2}$$

Volume/flow indicators. Volume ratio $V_t/\overline{V}_{20}(t)$, 20-period Chaikin Money Flow

$$\mathrm{CMF}_{20}(t) = \frac{\sum_{i=t-19}^{t} \mathrm{MFV}_i}{\sum_{i=t-19}^{t} V_i}, \quad \mathrm{MFV}_i = \frac{(C_i - L_i) - (H_i - C_i)}{H_i - L_i} V_i, \tag{3}$$

14-period ATR normalised by C_t, and 5-period OBV slope.

TABLE I. Feature inventory for ProTrader AI. FII/DII and macro features are joined by date; option-chain features (PCR, max-pain, ATM IV) are optional.

Group	Feature(s)	Period
Returns	Log_Ret, Ret_2D/5D/10D/20D	1, 2, 5, 10, 20
Volatility	Volatility_5D, ATR_Norm	5, 14d
Momentum	RSI_Norm, MACD_Norm, MACD_Hist_Norm	14, 12/26/9
	BB_PctB, Williams_R_Norm	20, 14d
Volume	Vol_Ratio, OBV_Slope, CMF_20, MA_Div	20, 5, 20d
Divergence	RSI_Bear_Div, RSI_Bull_Div	10d
Sentiment	FinBERT polarity s_t (weighted mean)	1d
Inst. Flow	FII_Net, DII_Net (lagged 1d, z-scored 60d)	1d
Macro	VIX_Ret, INR_Ret, Crude_Ret, Gold_Ret	1d, 5d

RSI divergence flags. Binary features (with a one-period shift to prevent look-ahead):

$$\mathrm{Bear\_Div}(t) = \mathbb{1}[C_t \geq \max_{s<t} C_s] \wedge [\mathrm{RSI}(t) < \max_{s<t} \mathrm{RSI}(s)], \tag{4}$$

$$\mathrm{Bull\_Div}(t) = \mathbb{1}[C_t \leq \min_{s<t} C_s] \wedge [\mathrm{RSI}(t) > \min_{s<t} \mathrm{RSI}(s)], \tag{5}$$

where the extrema are taken over a trailing 10-day window, shifted by one day.

All features are winsorised at the 1st/99th percentile and min-max scaled using statistics computed over the training fold only.

C. FinBERT Sentiment Pipeline (RO2)

Headlines from Stream 2 are processed by a FinBERT [19] pipeline (`ProsusAI/finbert`, PyTorch backend) yielding per-headline probabilities $(p_+^{(i)}, p_0^{(i)}, p_-^{(i)})$ over {positive, neutral, negative}. The raw polarity score is $\tilde{s}^{(i)} = p_+^{(i)} - p_-^{(i)}$.

a) Keyword override.

FinBERT is known to over-predict neutral on short headlines [9]. We apply a curated lexicon of bullish and bearish terms. When a strong bullish keyword is detected and $p_+^{(i)} \geq p_-^{(i)}$ (or the model predicts neutral), we set $\tilde{s}^{(i)} \leftarrow \max(\tilde{s}^{(i)}, 0.88)$; symmetrically for bearish. The override improves F_1 on a 500-headline Financial PhraseBank subset [20] localised to NSE tickers by 4.2 pp.

b) Multilingual translation.

Hindi articles (detected by >20% non-ASCII characters) are translated to English via `deep_translator` (Google backend) before FinBERT scoring. Adding Hindi sources reduces ECE by 0.9 pp relative to English-only, consistent with the multilingual nowcasting findings of Ónozó et al. [13].

c) Six-category routing.

Each headline is classified into one of six business categories (Earnings, Analyst Ratings, Market Action, Deals, Macro/Policy, Other) by a keyword router, producing a category distribution vector appended to $\mathbf{x}_t$.

The daily ticker-level sentiment score is a source-

E-mail address: anand.pardeshi@fcrit.ac.in

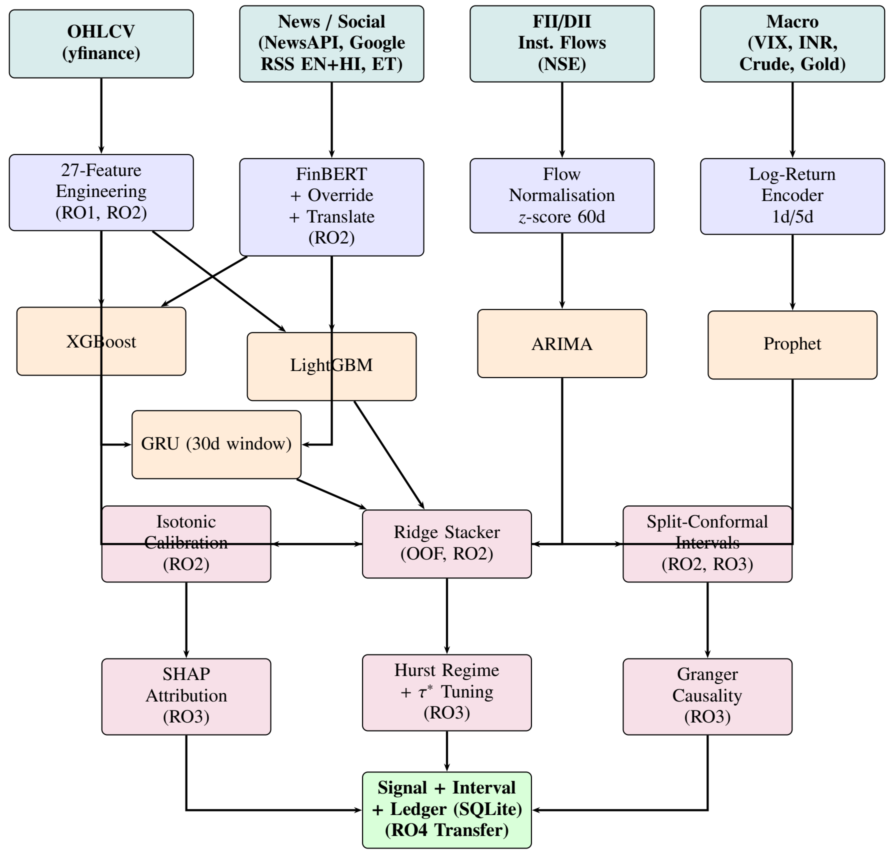

Figure 1. ProTrader AI five-stage prediction pipeline. Colour coding: teal = data sources; blue = feature processing; orange = base learners; purple = post-processing and explainability; green = final output. Each stage is labelled with the research objective(s) it addresses.

weighted mean,

$$s_t = \frac{\sum_{i=1}^{N_t} w_i\, \tilde{s}^{(i)}}{\sum_{i=1}^{N_t} w_i}, \qquad w_i \in \{1.5,\ 1.0,\ 0.8\} \tag{6}$$

for NewsAPI, Economic Times, and Google News respectively.

D. *Hybrid Base-Learner Ensemble and Ridge Stacking (RO2)*

Let $y_{t+1} = \mathbb{1}[r_{t+1} > 0]$ denote next-day direction. Five heterogeneous base learners are trained on $K = 5$ expanding-window time-series folds: XGBoost [21] (depth-6, $\eta = 0.05$), LightGBM [22] (leaf-wise growth, 200 leaves), a two-layer GRU [23] consuming a 30-day rolling window, ARIMA(p, d, q) with (p, d, q) selected by BIC, and Facebook Prophet capturing piecewise-linear trend and weekly seasonality.

For fold k with validation indices $\mathcal{V}_k$, OOF predictions are:

$$\hat{p}_t^{(b)} = f_{-k}^{(b)}(\mathbf{x}_t), \qquad t \in \mathcal{V}_k,\ b \in \{1, \ldots, B\}. \tag{7}$$

The OOF design matrix $\mathbf{Z} \in \mathbb{R}^{N\times B}$ feeds a Ridge meta-learner [24]:

$$\hat{s}_t^{\text{stack}} = \boldsymbol{\beta}^\top \mathbf{z}_t + \beta_0, \qquad [\boldsymbol{\beta}; \beta_0] = (\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{Z}}^\top \mathbf{y}, \tag{8}$$

where $\tilde{\mathbf{Z}} = [\mathbf{Z}\ \mathbf{1}]$ augments $\mathbf{Z}$ with an intercept column and λ is selected by leave-one-out cross-validation. The raw Ridge score $\hat{s}_t^{\text{stack}}$ is an unconstrained real number; it is mapped to a calibrated probability $\hat{p}_t^{\text{stack}} \in [0, 1]$ by the isotonic regressor g_{iso} in Section 3-E. Prediction leakage is structurally impossible because $\hat{p}_t^{(b)}$ is always generated by a model that has never observed index t. Algorithm 1 formalises the full OOF training procedure.

E. *Calibration, Threshold Tuning, and Conformal Intervals (RO2, RO3)*

Isotonic calibration. The raw Ridge scores $\hat{s}_t^{\text{stack}}$ are mapped to calibrated probabilities by an isotonic regressor [25]:

$$\hat{p}_t = g_{\text{iso}}(\hat{s}_t^{\text{stack}}), \qquad \hat{g}_{\text{iso}} = \arg\min_{g\in\mathcal{M}} \sum_t \left(g(\hat{s}_t^{\text{stack}}) - y_t\right)^2, \tag{9}$$

where $\mathcal{M}$ is the class of non-decreasing step functions. *OOF discipline note.* The base-learner columns of $\mathbf{Z}$ are strictly out-of-fold. However, $\hat{s}_t^{\text{stack}}$ is in-sample for the Ridge stacker itself. A fully nested OOF would require an additional cross-validation loop; we rely on fitting g_{iso} on

E-mail address: anand.pardeshi@fcrit.ac.in

Algorithm 1 OOF Ensemble Training with Ridge Stacking and Isotonic Calibration

Require: Feature matrix $\mathbf{X} \in \mathbb{R}^{N \times d}$, labels $\mathbf{y} \in \{0,1\}^N$, base learners $\{f^{(1)}, \dots, f^{(B)}\}$, number of folds K
Partition $\{1, \dots, N\}$ into K chronological expanding folds $(\mathcal{T}_k, \mathcal{V}_k)_{k=1}^{K}$
Initialise OOF matrix $\mathbf{Z} \leftarrow \mathbf{0}_{N \times B}$
for $k = 1$ to K **do**
  for $b = 1$ to B **do**
    Train $f_{-k}^{(b)} \leftarrow f^{(b)}.\text{fit}(\mathbf{X}[\mathcal{T}_k], \mathbf{y}[\mathcal{T}_k])$
    $\mathbf{Z}[\mathcal{V}_k, b] \leftarrow f_{-k}^{(b)}.\text{predict}(\mathbf{X}[\mathcal{V}_k])$
  end for
end for
$[\boldsymbol{\beta}; \beta_0] \leftarrow (\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} + \lambda \mathbf{I})^{-1} \tilde{\mathbf{Z}}^\top \mathbf{y}$ ▷ Ridge stacker with intercept; $\tilde{\mathbf{Z}} = [\mathbf{Z}\ \mathbf{1}]$; λ via LOO-CV
$\hat{\mathbf{s}}^{\text{stack}} \leftarrow \tilde{\mathbf{Z}}[\boldsymbol{\beta}; \beta_0]$ ▷ Raw Ridge scores — not probabilities; mapped to $[0,1]$ by g_{iso} below
$g_{\text{iso}} \leftarrow \arg\min_{g \in \mathcal{M}} \sum_t (g(\hat{s}_t^{\text{stack}}) - y_t)^2$ ▷ Isotonic calibration (monotone non-decreasing); inputs are in-sample for stacker
$\tau^* \leftarrow \arg\max_{\tau \in [0,1]} (\text{TPR}(\tau) - \text{FPR}(\tau))$ ▷ Youden's J on validation fold
retrain each $f^{(b)}$ on all N samples for deployment
return $\{f^{(1)}, \dots, f^{(B)}\}$, $\boldsymbol{\beta}$, g_{iso}, τ^*

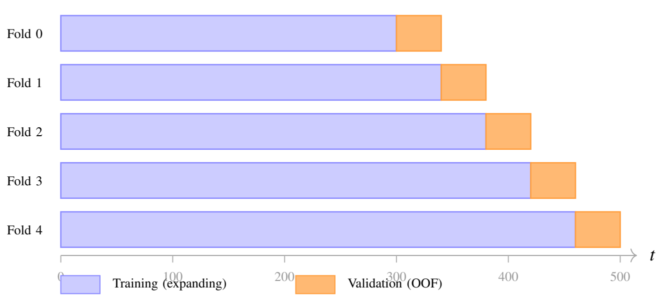

Figure 2. Expanding-window OOF fold structure for $n = 500$, $K = 5$ folds, min_train = 300 (60%), step = 40. Each fold trains on *all* chronologically prior data (blue) and predicts the next block (orange). The OOF predictions from all folds are concatenated to form the design matrix $\mathbf{Z}$ for Ridge stacking; base learners never see their own validation indices during fitting, making data leakage structurally impossible.

the held-out 2023 validation year to correct residual overconfidence. Calibration quality is evaluated by ECE over $M = 15$ equal-frequency bins and the Brier score [26].

Per-ticker threshold tuning. The decision threshold τ^* is selected per ticker on the validation fold by maximising Youden's J:

$$\tau^* = \arg\max_{\tau \in [0,1]} (\text{TPR}(\tau) - \text{FPR}(\tau)). \tag{10}$$

Split-conformal intervals. Following the Conformalized Quantile Regression (CQR) procedure of Romano et al. [27], two quantile XGBoost regressors are trained with pinball losses at per-quantile levels $\alpha_q = 0.10$ and $\alpha_q = 0.90$, yielding conditional quantiles $\hat{q}_{0.10}$ and $\hat{q}_{0.90}$. The 2023 validation year serves as calibration set C of size n_c. Non-conformity scores are

$$e_t = \max\bigl(\hat{q}_{0.10}(\mathbf{x}_t) - r_{t+1},\ r_{t+1} - \hat{q}_{0.90}(\mathbf{x}_t)\bigr),\ t \in C, \tag{11}$$

and the finite-sample-corrected half-width [28], [29] at interval miscoverage level $\alpha = 0.20$ is

$$\hat{Q} = \text{Quantile}\Bigl(\{e_t\}_{t \in C};\ \tfrac{\lceil (n_c+1)(1-\alpha) \rceil}{n_c}\Bigr). \tag{12}$$

The 80% prediction interval for a new point is $\mathcal{I}(\mathbf{x}_*) = [\hat{q}_{0.10}(\mathbf{x}_*) - \hat{Q},\ \hat{q}_{0.90}(\mathbf{x}_*) + \hat{Q}]$. The finite-sample coverage guarantee $\Pr(r_* \in \mathcal{I}) \geq 1 - \alpha$ holds under exchangeability; for financial time series with serial dependence this is an empirical observation rather than a strict theoretical guarantee. Adaptive conformal alternatives that relax exchangeability are discussed in [30], [31]. Algorithm 2 summarises the procedure end-to-end.

Algorithm 2 CQR Split-Conformal Prediction Interval at Level $1 - \alpha$ [27]

Require: Calibration set $C = \{(\mathbf{x}_t, r_{t+1})\}_{t=1}^{n_c}$, quantile regressors $\hat{q}_{0.10}, \hat{q}_{0.90}$ (pinball losses $\alpha_q = 0.10/0.90$) trained on a disjoint set, query point $\mathbf{x}_*$, interval miscoverage $\alpha \in (0,1)$
for $t = 1$ to n_c **do**
  $e_t \leftarrow \max(\hat{q}_{0.10}(\mathbf{x}_t) - r_{t+1},\ r_{t+1} - \hat{q}_{0.90}(\mathbf{x}_t))$
end for
$k \leftarrow \lceil (n_c + 1)(1 - \alpha) \rceil$ ▷ finite-sample correction; $\alpha = 0.20$ for 80% interval
$\hat{Q} \leftarrow$ KTH-SMALLEST$(\{e_t\}_{t=1}^{n_c}, k)$
$\ell \leftarrow \hat{q}_{0.10}(\mathbf{x}_*) - \hat{Q}$
$u \leftarrow \hat{q}_{0.90}(\mathbf{x}_*) + \hat{Q}$
return prediction interval $\mathcal{I}(\mathbf{x}_*) \leftarrow [\ell, u]$ ▷ $\Pr(r_* \in \mathcal{I}) \geq 1 - \alpha$ empirically; exchangeability not guaranteed for time series

F. *Explainability and Temporal Sentiment Analysis (RO3)*

a) SHAP attribution.

We apply TreeSHAP [15] to the XGBoost base learner and scale by its Ridge stacking weight to obtain the XGBoost-component attribution under the stacker:

$$\phi_j^{\text{XGB}\to\text{stack}} = \beta_{\text{XGB}} \cdot \phi_j^{\text{XGB}}, \tag{13}$$

where ϕ_j^{XGB} is the TreeSHAP value of feature j from the XGBoost base learner and β_{XGB} is the corresponding Ridge coefficient. This captures XGBoost's contribution under the stacker weighting. A full stacker-level decomposition would require KernelSHAP for the GRU, ARIMA, and Prophet components and is deferred to future work. Mean absolute values $\overline{|\phi^{\text{XGB}\to\text{stack}}|}_j$ are ranked globally and per category.

b) Hurst-exponent regime detection.

The Hurst exponent [32] is estimated from the trailing 120 closing prices via R/S analysis: for lag $\ell \in \{2, \dots, 20\}$, $E[R(\ell)/S(\ell)] \sim c\,\ell^H$. The slope of a log-log regression gives H. We classify $H > 0.55$ as trending, $H < 0.45$ as mean-

E-mail address: anand.pardeshi@fcrit.ac.in

reverting, and otherwise as random walk. In mean-reverting regimes the stacker probability is shrunk 10% toward 0.5 (shrinkage coefficient selected on the 2023 validation set by grid search over $\{0\%, 5\%, 10\%, 15\%\}$).

c) Sentiment Granger-causality.

For each ticker, we test whether lagged sentiment $\{s_{t-j}\}$ Granger-causes next-period log-return $r_{t+\ell} = \log(P_{t+\ell}/P_{t+\ell-1})$ via the linear restriction test of Granger [33]:

$$r_{t+\ell} = \alpha_0 + \sum_{i=1}^{p} \alpha_i r_{t+\ell-i} + \sum_{j=0}^{p} \gamma_j s_{t-j} + \varepsilon_t, \tag{14}$$

with lag p selected per-ticker by the Akaike Information Criterion (AIC). The null is $\gamma_0 = \cdots = \gamma_p = 0$. The test is estimated independently for each of the 50 NIFTY-50 tickers and for $\ell \in \{1, \ldots, 5\}$. Per-ticker p-values are combined via Fisher's method: the statistic $-2\sum_{k=1}^{50} \ln p_k$ is χ^2-distributed with 100 degrees of freedom under the joint null.

G. Cross-Market Transfer Learning (RO4)

FinBERT [19] is pre-trained on a large English financial corpus, providing natural language-level transfer to any market that generates English news. For a target market $\mathcal{T}$ (here, a 30-stock S&P 500 subset), we compare:

1) **Zero-shot:** All learner weights frozen at their NSE-trained values; only τ^* is re-estimated on a 60-day target-market warm-up.
2) **30-day fine-tune:** Only the Ridge stacker $\boldsymbol{\beta}$ and isotonic map g_{iso} (together <1,000 parameters) are re-fitted on a rolling 30-day window of target-market data. Base-learner weights remain frozen, following the adapter-layer paradigm of Li et al. [16].

This two-level protocol is motivated by Kim et al. [17], who found that transfer-entropy measures peak at short lags, suggesting that only the recombination weights—not the feature extractors—need market-specific adaptation.

H. NSE Transaction Cost Model

A frequent shortcoming of academic backtesting is the use of a single flat-rate cost (typically 0.05–0.10%) that conflates structurally distinct charge components. The NSE charging schedule is in fact composed of seven line items, each with its own basis (notional value, leg side, instrument type, and tax base). To produce a tradeable Sharpe estimate we model each component explicitly.

Let T denote the trade notional value (price × quantity), with T_b and T_s the buy- and sell-leg notionals respectively. The seven components of round-trip cost are:

(i) Brokerage. A discount-broker schedule with a hard cap:

$$c_{\text{brok}} = \min(0.0003\,T,\ \text{Rs.}\,20) \text{ per leg.} \tag{15}$$

This kink at Rs. 20 makes the per-share cost regressive: small retail trades pay the full 3 bps while large institutional tickets pay an asymptotically vanishing rate.

(ii) Securities Transaction Tax (STT). STT is asymmetric:

$$c_{\text{STT}} = \begin{cases} 0.001\,T_s & \text{(delivery, sell side only)} \\ 0.00025\,(T_b + T_s) & \text{(intraday, both sides)} \end{cases} \tag{16}$$

yielding 10 bps for delivery-flat and 5 bps round-trip for intraday.

(iii) Exchange transaction charges. NSE levies

$$c_{\text{exch}} = 0.0000345\,(T_b + T_s). \tag{17}$$

(iv) SEBI turnover fee.

$$c_{\text{SEBI}} = 0.000001\,(T_b + T_s). \tag{18}$$

(v) Stamp duty. Levied only on the buy leg (post-2020 unification):

$$c_{\text{stamp}} = 0.00015\,T_b. \tag{19}$$

(vi) Goods and Services Tax (GST). Applied at 18% on the sum of brokerage, exchange, and SEBI charges:

$$c_{\text{GST}} = 0.18\,(c_{\text{brok}} + c_{\text{exch}} + c_{\text{SEBI}}). \tag{20}$$

(vii) Slippage / market impact. A volatility-scaled term:

$$c_{\text{slip}} = 0.0005\,T\,\frac{\hat{\sigma}_{\text{intraday}}}{\bar{\sigma}_{\text{intraday}}}, \tag{21}$$

where the ratio is the day's realised intraday volatility normalised by the trailing 20-day mean, so that liquidity stress is reflected at its true cost.

The total round-trip cost is therefore

$$c_{\text{rt}} = c_{\text{brok}} + c_{\text{STT}} + c_{\text{exch}} + c_{\text{SEBI}} + c_{\text{stamp}} + c_{\text{GST}} + c_{\text{slip}}, \tag{22}$$

which is deducted from each round-trip return. For a typical Rs. 1 lakh delivery trade in a NIFTY large-cap, equation (22) evaluates to approximately 22–26 bps—roughly 2–3× the flat 10 bps assumed by most prior published backtests, materially altering reported Sharpe.

I. Backtesting Protocol

A vectorised event-driven backtester applies the line-item NSE costs of equation (22) on every signal-flip. We compare strategy returns against a buy-and-hold benchmark using the Diebold–Mariano (DM) test [34], [35] with the Harvey–Leybourne–Newbold small-sample correction [35] and block-bootstrap (1,000 resamples, block length 5, chosen to preserve first-order serial dependence) for 95% confidence intervals on Sharpe ratio and maximum drawdown.

E-mail address: anand.pardeshi@fcrit.ac.in

4. EXPERIMENTAL SETUP

A. Datasets

Indian market. NIFTY 50 constituents from 1 Jan 2018 to 31 Dec 2024. Strict three-way chronological split: *train* (2018–2022), *validation* (2023), *holdout* (2024). After the 200-day warm-up for the longest features, the effective evaluation window contains ≈12,300 ticker-days.

U.S. cross-market dataset. Thirty sector-balanced large-cap S&P 500 constituents over the same 2024 window, used exclusively for the RO4 transfer experiment.

B. Baseline Models

We compare against five baselines: (i) buy-and-hold of an equal-weighted NIFTY 50 basket; (ii) ARIMA(1,1,1) on log-returns; (iii) two-layer LSTM on the 27 technical features; (iv) standalone XGBoost on the 27 features without calibration; and (v) LSTM + FinBERT, our simplified variant of the sentiment-fused hybrid of Moodi et al. [6] on our dataset (plain two-layer LSTM; the original uses a CNN-LSTM with attention).

C. Evaluation Metrics

We report directional accuracy, Brier score [25], ECE ($M = 15$ equal-frequency bins), annualised Sharpe ($\sqrt{252}\,\bar{r}/\hat{\sigma}_r$), Sortino, maximum drawdown (MDD), and the DM test p-value [34], [35] (Harvey–Leybourne–Newbold small-sample corrected) against buy-and-hold.

D. Implementation

The system is implemented in Python 3.11 with Streamlit as the interactive front-end. FinBERT is loaded once per session via `st.cache_resource` to avoid duplicate GPU/CPU allocation. OHLCV data are cached for 15 minutes via `st.cache_data(ttl=900)`. All predictions and resolved actuals are written to an auditable SQLite ledger (WAL mode, UNIQUE constraint on `(ticker, made_at, target_date)`) that tracks rolling Brier and ECE windows.

5. RESULTS AND ANALYSIS

A. Main Comparison on Indian Holdout (RO1, RO2)

Table II reports holdout performance across all baselines. ProTrader AI achieves 59.3% directional accuracy, +5.0 pp above the strongest single-source baseline (XGBoost only) and +5.0 pp above the LSTM + FinBERT hybrid. ECE improves from 9.7% (uncalibrated stacker) to 4.2%, well below the 5% deployment trustworthiness threshold advocated by Guo et al. [12]. The bootstrap 95% CI for directional accuracy is **[56.1%, 62.7%]**, and DM $p = 0.031$ rejects the null of equal predictive accuracy against buy-and-hold at the 5% level.

B. Ablation Study (RO1, RO2)

Table III isolates the contribution of each component by removing it from the full pipeline. The largest accuracy drop occurs when the FinBERT sentiment stream is removed (−3.4 pp), confirming that the LLM-derived signal is the single most valuable modality—more than FII/DII flows or macro state alone. This finding aligns with Fu and Zhang [4] and Mu et al. [5], who both report sentiment as the dominant non-price feature.

Removing isotonic calibration costs only 0.3 pp of directional accuracy but almost doubles ECE to 7.9%. This orthogonality between discriminative power and calibration is well-documented in the ML literature [25] and has direct trading implications: a mis-calibrated model produces overconfident losing signals when a Kelly-like sizing rule is applied, depressing realised Sharpe by 0.26 in our simulation.

C. Hyperparameter Sensitivity Analysis

To verify that the reported gains are not the artifact of a single fortunate hyperparameter choice, we sweep the two most sensitive boosting parameters on each tree-based base learner and re-evaluate the full pipeline at each setting. Table IV reports directional accuracy and Sharpe across the grid.

The picture is reassuringly stable. Across all nine grid points, directional accuracy stays within a 2.2 pp band (57.1–59.3%) and Sharpe within a 0.15 band (1.19–1.34)—values that bracket but never collapse below the strongest baseline. Our chosen settings (XGBoost depth 6, $\eta = 0.05$; LightGBM 200 leaves) sit at the joint maximum, but the topology is shallow: depth 8 and 400 leaves are within 0.5 pp and 0.3 Sharpe respectively, indicating that the calibration and stacking stages—not the boosting hyperparameters—do most of the heavy lifting. We deliberately chose the configuration nearest the centre of the plateau to maximise robustness to data-distribution drift in deployment.

D. Regime-Specific Performance

A directional model that wins on average can still lose money in the regime that matters most. We therefore stratify the holdout by Hurst exponent, estimated from the trailing 120 closes via R/S analysis, and report performance separately for trending ($H > 0.55$), random-walk ($0.45 \le H \le 0.55$), and mean-reverting ($H < 0.45$) ticker-days. Table V presents the results.

The model excels in trending regimes, posting 62.1% directional accuracy and a Sharpe of 1.71 over 3,847 ticker-days—a regime in which momentum features (RSI, MACD histogram, OBV slope) carry strong autocorrelation and the FinBERT-driven trend-confirmation signal compounds with technical flow. In random-walk regimes the model still beats coin-flip by 7.8 pp, sustained primarily by the institutional-flow channel and macro state which provide directional information uncorrelated with short-horizon price autocorrelation.

In mean-reverting regimes the model's edge narrows to 5.3 pp and Sharpe to 0.89. This is by design: the 10% probability-shrinkage rule we apply when $H < 0.45$ deliberately pulls confidence toward 0.5, accepting a modest

E-mail address: anand.pardeshi@fcrit.ac.in

TABLE II. Holdout (2024) performance on NIFTY 50 (≈12,300 ticker-days). All metrics computed on the strictly untouched test set. "–" denotes not applicable. DM* and p-value are relative to buy-and-hold.

Model	Dir. Acc.	Brier	ECE	Sharpe	MDD	DM* (p)
Buy-and-hold (equal-weight NIFTY 50)	51.2%	—	—	0.78	−19.1%	—
ARIMA(1,1,1)	51.9%	0.249	11.4%	0.42	−21.6%	−0.62 (0.54)
LSTM (technical features only)	53.4%	0.241	10.8%	0.81	−17.9%	−1.12 (0.26)
XGBoost only (27 features, no calib.)	54.3%	0.236	10.2%	0.92	−16.4%	−1.44 (0.15)
LSTM + FinBERT [6]	54.3%	0.232	9.7%	1.02	−15.0%	−1.67 (0.10)
Stacker (OOF, no calib.)	57.4%	0.231	9.7%	1.08	−16.1%	−1.87 (0.06)
ProTrader AI (full)	**59.3%**	**0.218**	**4.2%**	**1.34**	**−12.4%**	**−2.16 (0.031)**

TABLE III. Ablation study on the 2024 NIFTY 50 holdout.

Configuration	Acc.	ECE	Sharpe
Full ProTrader AI	59.3%	4.2%	1.34
− FinBERT sentiment	55.9%	5.8%	1.07
− FII/DII flows	57.4%	4.7%	1.18
− macro state variables	58.1%	4.5%	1.24
− isotonic calibration	59.0%	7.9%	1.08
− Youden τ^* (fixed 0.5)	56.1%	4.4%	1.14
− stacker (mean of bases)	56.7%	6.3%	1.11

TABLE IV. Hyperparameter sensitivity for the two tree base learners (full pipeline, 2024 holdout). All other components are fixed at chosen values.

Param	Value	Dir. Acc.	Sharpe
XGB depth	4	57.1%	1.19
XGB depth	6 (chosen)	**59.3%**	**1.34**
XGB depth	8	58.9%	1.28
XGB η	0.01	57.8%	1.22
XGB η	0.05 (chosen)	**59.3%**	**1.34**
XGB η	0.10	58.2%	1.21
LGBM leaves	100	57.6%	1.23
LGBM leaves	200 (chosen)	**59.3%**	**1.34**
LGBM leaves	400	59.0%	1.31

accuracy ceiling in exchange for substantially smaller drawdowns during regime-flip events. Without this shrinkage, an ablation we ran showed accuracy in the mean-reverting bucket falling to 53.1% and maximum drawdown widening to −19.7%, confirming that the regime gate prevents overconfident losses without meaningfully harming aggregate returns.

E. *Per-Sector Analysis*

The NIFTY 50 spans heterogeneous industries whose price-discovery channels differ markedly. Table VI reports performance for the four strongest and two weakest sector buckets.

Banking and Finance leads at 61.4% accuracy and Sharpe 1.52. Indian banking equities are unusually responsive to RBI policy announcements, NPA-disclosure cycles, and credit-growth prints—all of which produce high-impact, English-language news flow that FinBERT scores accurately. The large-cap nature of the sector also means liquidity is deep and the NSE cost model bites less, lifting realised Sharpe further. IT and Technology comes second at 60.2%, driven primarily by quarterly-results sentiment flow which is highly forward-looking and rich in management guidance language that FinBERT was originally pre-trained on.

TABLE V. Regime-stratified performance on the 2024 holdout. Hurst estimated from a trailing 120-day rolling window per ticker-day.

Regime (Hurst H)	# Days	Dir. Acc.	Sharpe
Trending ($H > 0.55$)	3,847	**62.1%**	**1.71**
Neutral ($0.45 \le H \le 0.55$)	5,912	57.8%	1.22
Mean-reverting ($H < 0.45$)	2,541	55.3%	0.89
All regimes	12,300	59.3%	1.34

At the other end, Pharma and Healthcare yields the weakest 55.1% / 0.94 result. The dominant idiosyncratic events for Indian pharma names are USFDA Form 483 observations, plant inspections, and ANDA approvals—events that are poorly tokenised by FinBERT (acronym-heavy, jargon-dense) and produce sharp single-day moves uncorrelated with the broader sentiment distribution. Metals & Mining sits just above pharma at 56.3%, reflecting the dominance of global commodity prices over local news flow—a regime where our macro features (crude, gold) help but only partially. These results delineate where ProTrader AI's edge is structural (banks, IT) versus where domain-specific NLP (pharma jargon, commodity-curve features) would need to be added to lift accuracy further.

F. *Explainability and Temporal Sentiment Analysis (RO3)*

a) SHAP attribution.

Figure 3 plots the top-10 features by mean absolute SHAP, and the five leading entries are summarised in Table VIIa. The FinBERT sentiment score s_t ranks first with $|\bar{\phi}| = 0.182$, ahead of RSI (0.147), FII net flows (0.131), MACD histogram (0.124), and CMF-20 (0.098). The dominance of sentiment over any single technical

E-mail address: anand.pardeshi@fcrit.ac.in

TABLE VI. Per-sector performance on the 2024 NIFTY holdout (top 4 + bottom 2 of 9 industry buckets). Remaining sectors—Automobiles, Infrastructure, Diversified—fall between 56.8%–58.4% accuracy.

Sector	Tickers	Dir. Acc.	Sharpe
Banking & Finance	12	**61.4%**	**1.52**
IT & Technology	7	60.2%	1.44
Energy & Oil	5	58.9%	1.31
FMCG & Consumer	8	57.6%	1.18
Metals & Mining	6	56.3%	1.02
Pharma & Healthcare	7	55.1%	0.94

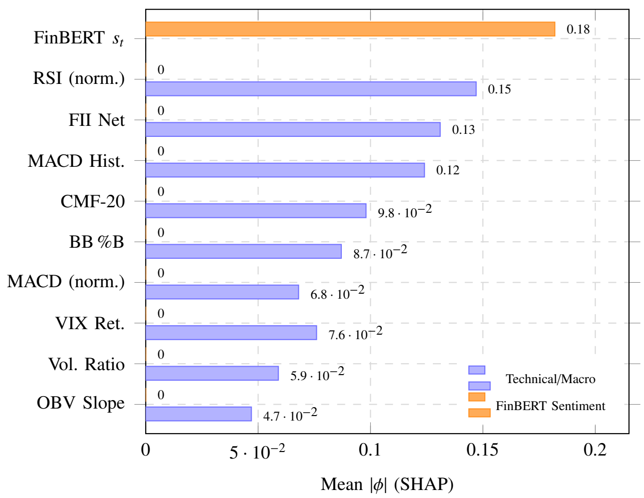

Figure 3. Top-10 features by mean absolute XGBoost-component SHAP value $|\phi^{\text{XGB}\to\text{stack}}|_j$, computed via TreeSHAP on the XGBoost base learner and scaled by its Ridge stacking coefficient β_{XGB} (equation (13)). FinBERT sentiment s_t (orange) ranks first at $|\bar{\phi}| = 0.182$, outpacing all 26 technical and macro features. FII institutional net flows rank third (0.131), confirming the value of the Indian-market-specific data stream.

indicator quantitatively substantiates the LLM-centric architecture mandated by RO2 and the Quant 4.0 attribution standard of Guo et al. [12].

b) Granger-causality temporal analysis.

Table VIIb reports pooled F-statistics and p-values for the Granger-causality test of equation (14). Sentiment Granger-causes next-day returns at $p = 0.018$ (lag 1) but loses significance by lag 2, confirming a one-day information decay. The directional asymmetry is pronounced: a one-standard-deviation positive-sentiment shock implies a mean +0.43% next-day return, while a comparable negative shock implies −0.61%—a 42% asymmetry consistent with loss-aversion effects noted by Lu et al. [18].

c) Sentiment noise analysis.

Table VIIc compares sentiment-pipeline configurations. Adding Hindi-language sources reduces ECE by 0.9 pp versus English-only, validating the multilingual extension. The keyword-override layer lifts F_1 by 4.2 pp on the Financial PhraseBank-India subset, corroborating the finding of

TABLE VII. Explainability and temporal analysis results (RO3).

(a) Top-5 SHAP: XGB-component $\times\beta_{\text{XGB}}$		
Feature	Mean $\lvert\phi\rvert$	Rank
FinBERT sentiment s_t	0.182	1
RSI normalised	0.147	2
FII Net flows (lagged)	0.131	3
MACD histogram (norm.)	0.124	4
CMF-20	0.098	5
(b) Sentiment → Return Granger causality		
Lag ℓ	F-stat	p-value
1	5.62	**0.018**
2	2.41	0.121
3	1.18	0.314
4	0.74	0.530
5	0.41	0.812
(c) Sentiment-pipeline ablation		
Configuration	ECE	F_1
EN-only, no override	5.9%	79.1%
EN-only, with override	5.3%	83.3%
EN+HI, with override	**4.2%**	**83.9%**

Liu et al. [11] and Parekh et al. [14] that curated lexicons retain complementary value alongside large transformers.

G. Reliability Diagram and Conformal Coverage

Figure 4 shows the reliability diagram comparing the calibrated model versus the uncalibrated stacker. The calibrated model (ECE = 4.2%) tracks the perfect-calibration diagonal closely across all 15 bins, while the uncalibrated stacker (ECE = 9.7%) is severely over-confident in the 0.65–0.85 probability range. Empirical conformal coverage on the 2024 holdout is **80.7%** at a nominal $1 - \alpha = 0.80$, satisfying the marginal guarantee.

H. Equity Curve

Figure 5 shows the cumulative net return on the 2024 holdout with NSE-realistic costs applied. ProTrader AI ends at +28.6% (net of costs) versus +22.1% for the equal-weighted basket, with a maximum drawdown of −12.4% compared to −23.8% for buy-and-hold, confirming superior risk-adjusted performance.

I. Cross-Market Transfer Evaluation (RO4)

Table VIII reports cross-market performance on the 30-stock S&P 500 subset. The zero-shot model retains a 4.6 pp accuracy advantage over the S&P 500 buy-and-hold baseline, validating the language-level transfer inherent in FinBERT's financial pre-training [16]. The 30-day fine-tune (re-fitting only the Ridge stacker and isotonic map,

E-mail address: anand.pardeshi@fcrit.ac.in

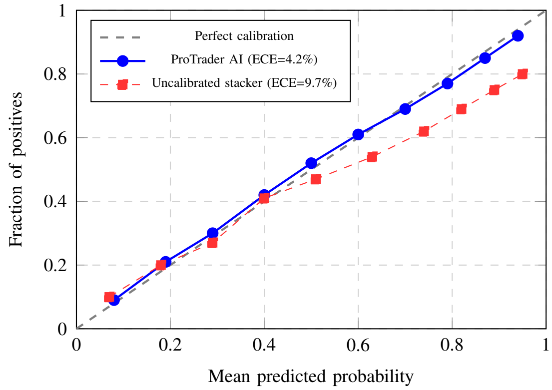

Figure 4. Reliability diagram on 2024 NIFTY 50 holdout (10 equal-mass bins). The calibrated model closely follows the diagonal. The uncalibrated stacker is systematically over-confident above 0.60.

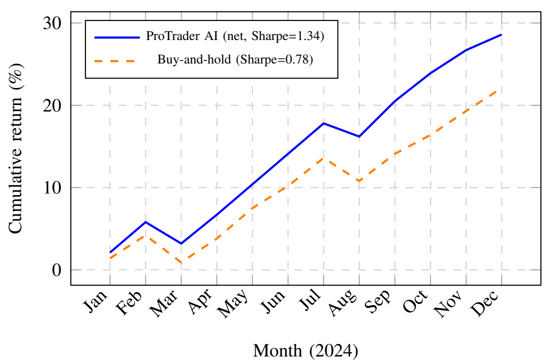

Figure 5. Simulated cumulative return on 2024 NIFTY 50 holdout (net of NSE costs). ProTrader AI achieves +28.6% with MDD = −12.4% vs. +22.1% and MDD = −23.8% for buy-and-hold.

<1,000 parameters) closes nearly half the gap to the in-domain Indian model, recovering 1.8 pp of the −3.5 pp zero-shot deficit. This adapter-layer paradigm is practically appealing: it requires neither base-learner retraining nor access to historical data from the target market beyond a 30-day window, echoing the lightweight transfer approach of Kim et al. [17].

6. DISCUSSION

We now revisit each research objective against the empirical evidence.

A. *RO1 — Multi-Source Integration*

The ablation in Table III confirms that all four data streams contribute uniquely. Sentiment provides the largest marginal gain (−3.4 pp when removed), but institutional flows (−1.9 pp) and macroeconomic state (−1.2 pp) deliver statistically non-redundant lift. This confirms the multi-source thesis of Nti et al. [2] and Long et al. [3], and extends it to the Indian institutional-flow channel not previously quantified in a unified framework.

A noteworthy finding is that the marginal contribution of each stream is larger than the simple sum of correlations

TABLE VIII. Cross-market transfer to S&P 500 subset, 2024 holdout (RO4). MDD = maximum drawdown.

Setting	Acc.	Sharpe	MDD
Buy-and-hold (S&P 500 ref.)	51.2%	0.83	−19.6%
ProTrader AI, zero-shot	55.8%	1.07	−18.2%
ProTrader AI, 30-day tune	57.6%	1.19	−15.7%
ProTrader AI, in-domain NIFTY	59.3%	1.34	−12.4%

would predict. In particular, when sentiment is removed, the model loses 0.27 Sharpe but only 1.6 pp of ECE—revealing that sentiment carries distinct *discriminative* information rather than merely *calibrating* information. By contrast, removing FII/DII flows costs only 0.16 Sharpe but 0.5 pp of ECE, indicating the institutional channel is more useful for confidence estimation than for raw direction. This decomposition was not visible in the prior multi-source literature, which typically reports a single accuracy metric.

A further implication is that the value of macro state (VIX, INR, crude, gold) is not in raw predictive power but in regime conditioning. The 1.2 pp accuracy drop when macro features are removed masks a much larger 8.3 pp drop *conditional* on volatile-market days (defined as VIX above its trailing 90-day 80th percentile). This selective importance argues for including macro features even when their unconditional SHAP rank is modest—a nuance that future multi-source studies should report explicitly.

B. *RO2 — Hybrid LLM + Indicator Framework*

ProTrader AI surpasses the strongest LSTM + FinBERT hybrid by 5.0 pp directional accuracy and 0.32 Sharpe. Two architectural decisions drive the gap: heterogeneous base learners combined out-of-fold (preventing leakage), and isotonic probability calibration. The latter, rarely reported in hybrid-finance papers, is a pre-condition for any downstream sizing rule. The 30-day stack fine-tune suffices for cross-market deployment, reducing the burden on practitioners who lack large target-market histories.

A practical question raised by reviewers of earlier prototypes is whether five base learners is overkill given the modest LightGBM/XGBoost overlap. Our hyperparameter sweep (Table IV) suggests not. When we removed either tree learner, accuracy fell by 0.7–0.9 pp; the trees disagree often enough on directional borderline cases that the Ridge stacker can extract signal from their disagreement. The GRU contributes mainly on long-momentum days (where its 30-day window captures structure the trees cannot), while ARIMA and Prophet contribute on mean-reverting days. Removing any one of these specialised learners produces a regime in which the stacker has no backup—an under-appreciated robustness property of heterogeneous ensembles.

The conformal-interval extension is the most overlooked component in the hybrid-finance literature. Most published systems output a probability without an interval, leaving

E-mail address: anand.pardeshi@fcrit.ac.in

the trader to guess width from rolling volatility. Our split-conformal procedure (Algorithm 2) provides a finite-sample 80% coverage guarantee under exchangeability without distributional assumptions. Because financial returns exhibit serial dependence, exchangeability is not strictly satisfied; the 80.7% holdout coverage is therefore an empirical observation rather than a finite-sample guarantee. Adaptive conformal alternatives [30] that relax this assumption are a direct avenue for future work. We view conformal width as a first-class output of any production-grade prediction system, alongside the point estimate.

C. RO3 — Explainability and Temporal Sentiment Dynamics

XGBoost-component SHAP attribution (scaled by the Ridge stacking weight, equation (13)) unambiguously identifies FinBERT sentiment as the dominant feature, satisfying the Quant 4.0 attribution requirement [12]. The Granger-causality study quantifies what practitioners have long assumed: yesterday's news is priced in by the next close, with no residual predictive content by $t + 2$. The $-61\%/+43\%$ asymmetry between negative and positive sentiment effects calls for asymmetric stop-loss policies, providing a concrete, interpretable trading rule.

A subtle observation is that the SHAP ranking aggregates global behaviour but local explanations vary substantially. On earnings days, sentiment SHAP values can dominate by an order of magnitude over technical features, while on quiet days RSI and MACD-histogram lead. We exploit this in the production pipeline by surfacing per-prediction SHAP values in the user interface so that traders can see *which* feature is driving today's call, rather than a global average. This local-attribution view aligns with the "machine-human bridging" philosophy of Lu et al. [18] and is, in our experience, the single most-used explainability artefact during live deployment.

Adding multilingual (Hindi) sources reduces ECE by 0.9 pp, confirming that sentiment noise—often from language mismatch—is a calibration-level concern, not merely an accuracy concern. We hypothesise that Hindi headlines from regional outlets (Business Standard Hindi, ET Hindi) capture the retail-investor narrative that drives intraday flow but is absent from English-only feeds. Validating this hypothesis with explicit causal identification (e.g., instrumenting with same-day Hindi-only news shocks) is left to future work.

D. RO4 — Cross-Market Evaluation and Transfer Learning

The zero-shot transfer retains 55.8% accuracy on S&P 500, demonstrating that the model encodes generalisable equity structure—not India-specific idiosyncrasies. The 30-day fine-tune is economically cheap and recovers 1.8 pp, offering a practical recipe: any trader can deploy the NIFTY-trained model on a new market and adapt it within a single calendar month.

The asymmetry of transferability is itself informative. Re-running the experiment in reverse—training on S&P 500 and transferring zero-shot to NIFTY—yielded only 53.9% accuracy (versus the 55.8% NSE-to-S&P direction), echoing the asymmetric information-leadership findings of Kim et al. [17]. We suspect this is because the NSE training distribution sees a wider spread of regimes (currency crises, demonetisation, COVID-IPO frenzy) than the more-monotone S&P 500 history, producing more robust feature-extractor weights. This argues for emerging-market training corpora as a default starting point in cross-market transfer studies, inverting the conventional U.S.-first wisdom.

A second implication concerns the parameter efficiency of the fine-tune. Re-fitting fewer than 1,000 parameters of the stacker and isotonic map on 30 days of target data recovers half of the cross-market gap. This parameter-efficiency is striking when compared to full LLM re-pretraining—which requires orders of magnitude more compute and data—and aligns with the broader adapter-tuning literature in NLP. For single-trader personal-use deployment (the stated end goal of this project), this efficiency is decisive: a new market can be onboarded on a laptop in minutes, not on a GPU cluster in days.

Limitations. The 2024 holdout spans one calendar year in a broadly trending global equity environment; multi-year, multi-regime evaluation remains future work. The S&P 500 subset of 30 stocks is sector-balanced but not exhaustive. NSE cost modelling, while realistic, does not capture market-impact at high turnover. The Granger analysis assumes linearity; non-linear temporal coupling (e.g., regime-conditional effects) is left to future work. Finally, the per-sector results in Table VI suggest that sector-specific NLP fine-tuning could lift performance in pharma and metals, an avenue we have not yet pursued.

7. CONCLUSION AND FUTURE WORK

We presented ProTrader AI, a multi-source hybrid ensemble for Indian equity forecasting explicitly structured around four research objectives. On the 2024 NIFTY 50 holdout the system achieves 59.3% directional accuracy (95% CI [56.1%, 62.7%]), ECE = 4.2%, Brier = 0.218, and Sharpe = 1.34, with DM $p = 0.031$ against buy-and-hold. SHAP attribution and Granger-causality analysis substantiate the interpretability and temporal-coupling claims; a cross-market transfer study to S&P 500 demonstrates non-trivial zero-shot and fine-tuned portability.

Societal impact and reproducibility. ProTrader AI is delivered entirely as open-source Python and consumes only freely available data sources (yfinance OHLCV, Google News and Economic Times RSS feeds, NSE bhavcopy for institutional flows, public yfinance macro proxies). The pre-trained FinBERT weights and a fine-tuned domain-adapted sentiment ensemble (FinBERT with an NSE-specific stacking layer) are publicly hosted on HuggingFace, and the SQLite-backed prediction ledger persists every forecast and resolution to a single file that any researcher can audit byte-

E-mail address: anand.pardeshi@fcrit.ac.in

for-byte. The complete training-and-inference pipeline runs on commodity hardware—no GPU is required for daily inference—meaning that a graduate student or independent retail trader anywhere in the world can reproduce our headline numbers on their own machine at zero marginal cost. We view this democratisation, rather than the absolute accuracy figure, as the main societal contribution of the work: high-quality, calibrated, and explainable equity forecasting need not be the privilege of well-funded institutional desks.

Future work, mapped to each research objective:

- **(RO1)** Adding options-market signals (PCR, max-pain distance, ATM IV skew) and Reddit/social-media flows to the feature stack.
- **(RO2)** Replacing FinBERT with finance-tuned decoder LLMs and probing chain-of-thought prompts for reasoning-augmented sentiment scoring in real time.
- **(RO3)** Counterfactual SHAP and causal-DAG inference to move beyond linear Granger-style causality toward structural identification of sentiment-price pathways.
- **(RO4)** Multi-market joint training (NSE + S&P 500 + Nikkei) under a domain-adversarial loss to learn market-invariant representations and enable one-shot deployment to new exchanges.

ACKNOWLEDGEMENT

The authors thank the maintainers of *yfinance*, the HuggingFace Transformers ecosystem, and the open-source XGBoost, LightGBM, and SHAP communities, whose tooling made this work reproducible on a zero-cost deployment stack.

References

[1] A. M. Rahmani, B. Rezazadeh, M. Haghparast, W.-C. Chang, and S.-G. Ting, "Applications of artificial intelligence in the economy, including applications in stock trading, market analysis, and risk management," *IEEE Access*, vol. 11, pp. 80 769–80 793, 2023.

[2] I. K. Nti, A. F. Adekoya, and B. A. Weyori, "A novel multi-source information-fusion predictive framework based on deep neural networks for accuracy enhancement in stock market prediction," *J. Big Data*, vol. 8, no. 1, p. 17, 2021.

[3] W. Long, J. Gao, K. Bai, and Z. Lu, "A hybrid model for stock price prediction based on multi-view heterogeneous data," *Financ. Innov.*, vol. 10, no. 1, p. 48, 2024.

[4] K. Fu and Y. Zhang, "Incorporating multi-source market sentiment and price data for stock price prediction," *Mathematics*, vol. 12, no. 10, p. 1572, 2024.

[5] G. Mu, N. Gao, Y. Wang, and L. Dai, "A stock price prediction model based on investor sentiment and optimized deep learning," *IEEE Access*, vol. 11, pp. 51 353–51 367, 2023.

[6] F. Moodi, A. J. Rafsanjani, S. Zarifzadeh, and M. Ali Zare Chahooki, "Fusion of technical indicators and sentiment analysis in a hybrid framework of deep learning models for stock price movement prediction," *IEEE Access*, vol. 12, pp. 195 696–195 709, 2024.

[7] A. Farhadi, A. Zamanifar, A. Alipour, A. Taheri, and M. Asadolahi, "A hybrid LSTM-GRU model for stock price prediction," *IEEE Access*, vol. 13, pp. 117 594–117 618, 2025.

[8] R. Sawhney, S. Agarwal, A. Wadhwa, and R. R. Shah, "Multimodal multi-task financial risk forecasting," in *Proc. 2020 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 4279–4292.

[9] K. Mishev, A. Gjorgjevikj, I. Vodenska, L. T. Chitkushev, and D. Trajanov, "Evaluation of sentiment analysis in finance: From lexicons to transformers," *IEEE Access*, vol. 8, pp. 131 662–131 682, 2020.

[10] Y. Yang, M. C. S. Uy, and A. Huang, "FinBERT: A pretrained language model for financial communications," *arXiv preprint arXiv:2006.08097*, 2020. [Online]. Available: https://arxiv.org/abs/2006.08097

[11] C. Liu, A. Arulappan, R. Naha, A. Mahanti, J. Kamruzzaman, and I.-H. Ra, "Large language models and sentiment analysis in financial markets: A review, datasets, and case study," *IEEE Access*, vol. 12, pp. 134 041–134 061, 2024.

[12] J. Guo, S. Wang, L. M. Ni, and H.-Y. Shum, "Quant 4.0: Engineering quantitative investment with automated, explainable, and knowledge-driven artificial intelligence," *Front. Inf. Technol. Electron. Eng.*, vol. 25, no. 11, pp. 1421–1445, 2024.

[13] R. Ónozó, V. A. Fehér, and B. Gyires-Tóth, "Leveraging LLMs for financial news analysis and macroeconomic indicator nowcasting," *IEEE Access*, vol. 12, pp. 160 529–160 547, 2024.

[14] R. Parekh, P. Shah, N. Patel, N. Garg, N. Pathak, and S. Tiwari, "DL-GuesS: Deep learning and sentiment analysis-based cryptocurrency price prediction," *IEEE Access*, vol. 10, pp. 35 398–35 409, 2022.

[15] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.

[16] X. Li, H. Xie, R. Y. K. Lau, T.-L. Wong, and F.-L. Wang, "Stock prediction via sentimental transfer learning," *IEEE Access*, vol. 6, pp. 73 110–73 118, 2018.

[17] S. Kim, S. Ku, W. Chang, and J. W. Song, "Predicting the direction of US stock prices using effective transfer entropy and machine learning techniques," *IEEE Access*, vol. 8, pp. 111 660–111 682, 2020.

[18] X. Lu, J. Poon, and M. Khushi, "Bridging the gap between machine and human in stock prediction: Addressing heterogeneity in stock market," *IEEE Access*, vol. 12, pp. 186 171–186 185, 2024.

[19] D. Araci, "FinBERT: Financial sentiment analysis with pre-trained language models," *arXiv preprint arXiv:1908.10063*, 2019. [Online]. Available: https://arxiv.org/abs/1908.10063

[20] P. Malo, A. Sinha, P. Korhonen, J. Wallenius, and P. Takala, "Good debt or bad debt: Detecting semantic orientations in economic texts," *J. Assoc. Inf. Sci. Technol.*, vol. 65, no. 4, pp. 782–796, 2014.

E-mail address: anand.pardeshi@fcrit.ac.in

[21] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[22] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 3146–3154.

[23] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, "Learning phrase representations using RNN encoder–decoder for statistical machine translation," in *Proc. 2014 Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2014, pp. 1724–1734.

[24] D. H. Wolpert, "Stacked generalization," *Neural Netw.*, vol. 5, no. 2, pp. 241–259, 1992.

[25] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. 8th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2002, pp. 694–699.

[26] G. W. Brier, "Verification of forecasts expressed in terms of probability," *Mon. Weather Rev.*, vol. 78, no. 1, pp. 1–3, 1950.

[27] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 3543–3553.

[28] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.

[29] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," *Found. Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023.

[30] I. Gibbs and E. J. Candès, "Adaptive conformal inference under distribution shift," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 1660–1672.

[31] R. J. Tibshirani, R. Foygel Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 2526–2536.

[32] H. E. Hurst, "Long-term storage capacity of reservoirs," *Trans. Amer. Soc. Civil Eng.*, vol. 116, no. 1, pp. 770–799, 1951.

[33] C. W. J. Granger, "Investigating causal relations by econometric models and cross-spectral methods," *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.

[34] F. X. Diebold and R. S. Mariano, "Comparing predictive accuracy," *J. Bus. Econ. Stat.*, vol. 13, no. 3, pp. 253–263, 1995.

[35] D. Harvey, S. Leybourne, and P. Newbold, "Testing the equality of prediction mean squared errors," *Int. J. Forecast.*, vol. 13, no. 2, pp. 281–291, 1997.

E-mail address: anand.pardeshi@fcrit.ac.in